

Aaed Musa — Internal Cycloidal Actuator (및 관련 액추에이터 프로젝트)

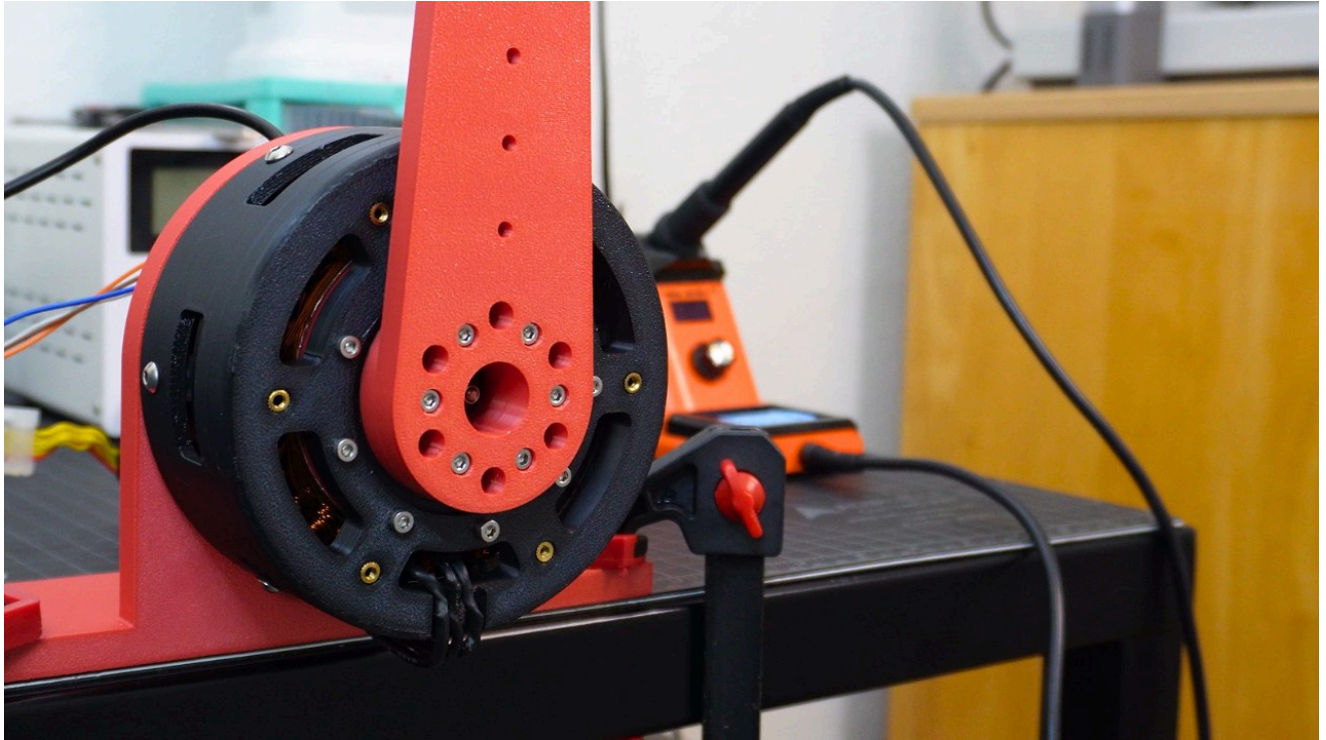
웹페이지 글·그림 발췌 — 2026-09-05 수집

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Internal Cycloidal Actuator

출처: <https://www.aaedmusa.com/projects/internalcycloidallactuator>

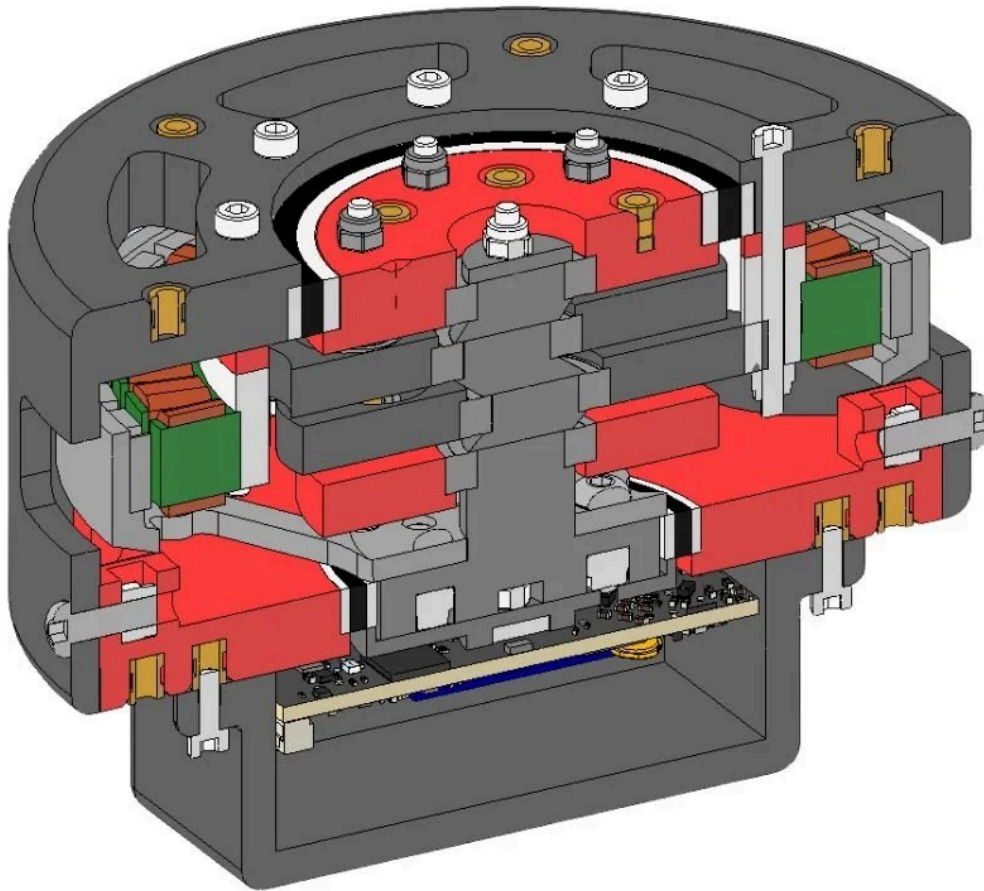


Internal Cycloidal Actuator (02/14/2024)

About

Having designed many robotic actuators, one recurring challenge that I found is achieving compactness. Ideally, a robotic actuator should be as slim as possible, especially for legged robots. To fix this, I decided to try a new design approach. Instead of using an off-the-shelf BLDC motor for the actuator, I decided to design my own BLDC motor. This gave me the liberty to add the gearbox in the center of the motor thus making the actuator compact. In my opinion, this is the optimal actuator design as there are no performance tradeoffs. I first saw this design scheme implemented in the MIT Mini Cheetah robot which has a 6:1 internal planetary gear drive in the center of its actuators.

- GitHub (For CAD and BOM)



Specs and Results

- 36N42P Configuration
- $\varnothing 125 \times 84\text{mm}$
- Total Mass: 1023g
- Total Cost: \$384
- Speed: 209RPM @ 22.2V
- Torque: 16.17Nm
- Motor phase resistance: 75 mOhm
- Motor phase inductance: 41.05 μH

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How it works

The actuator design utilizes the QDD (Quasi Direct Drive) scheme which means that it uses a low gear reduction to retain speed and efficiency. This also means that the motor needs to have high torque in order for the actuator to be powerful. I did this by designing an outrunner BLDC motor. Outrunner motors have high torque densities because of their large gap radii. I purchased a 10010 stator with 36 slots from Aliexpress. I coiled it myself using 6x strands of 26AWG copper wire doing 6 turns per slot. (effectively 36 turns per slot). The stator is made from laminated steel sheets which should increase the flux of the motor as well as prevent eddy current loss.

I designed the rotor of the motor and machined it out of mild steel 1045 using PCBWay. Making it out of steel should increase the flux of the permanent magnets attached to the rotor. I mounted 42x N52 grade magnets to the rotor using JB Weld. Interestingly JB Weld has 11% steel making it ferromagnetic. This should actually increase flux though not by a significant amount.

The actuator uses an 8:1 cycloidal gear drive which is located in the center of the motor. The Fixed ring of the gearbox (composed of roller pins) was machined out of aluminum 6061 and was fit into the center of the stator with Loctite 648 retaining compound. The eccentric shaft of the gearbox directly mounts to the rotor of the motor. The gearbox is essentially part of the motor. Designing the gearbox was the most challenging part of this project since it required the use of small bearings and spacers.

The actuator uses an ODrive S1 FOC Controller giving it the ability to lift loads and move to exact positions. Overall, this actuator worked quite well. The only issue that I ran into was with the 3D-printed parts slightly warping due to the heat from the coils. A fully metal gearbox is probably the best way to go next time.



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Rubik's Cube Solving Robot 2.0

OpenQDD Actuator

출처: <https://www.aaedmusa.com/projects/openqdd>



OpenQDD V1 (06/21/2023)

About

OpenQDD is a 3D-printed open-source robotic actuator that I designed to be a powerful, easy-to-build, and relatively inexpensive solution to creating dynamic robots. Robots like Spot, Atlas, and the MIT mini cheetah easily impress the eye with their agility, speed, and overall ability to mimic natural motion. With OpenQDD, creating high-performing actuators at an affordable price is possible.

- Building Instructions
- Github (For CAD, Code, and BOM)

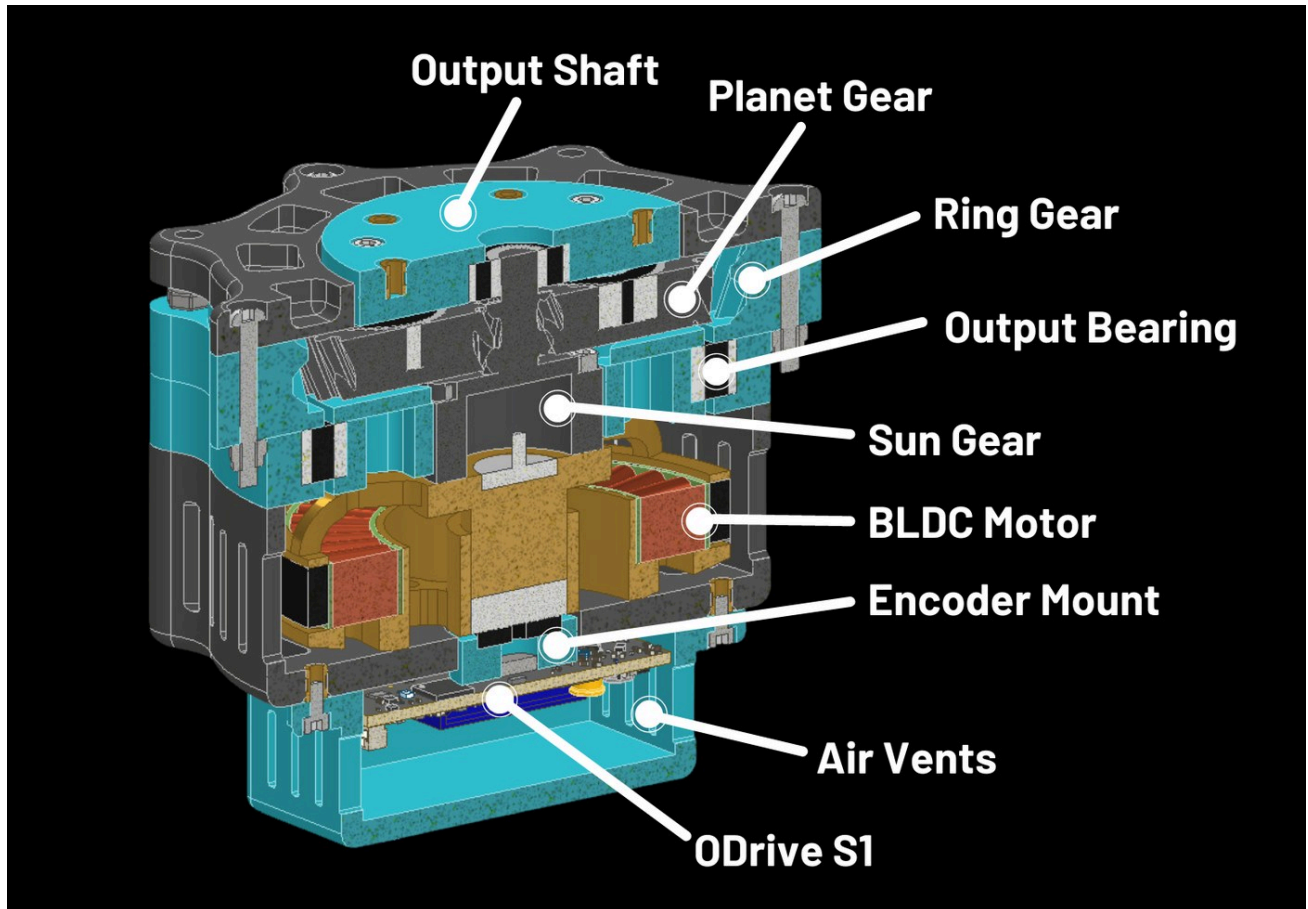
This project is heavily inspired by the OpenTorque actuator made by Gabrael Levine as well as the Cycloidal Actuator made by James Bruton. These open-source actuators follow the same working principle as OpenQDD and are also fully 3D printed.

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Specifications

- 9:1 Planetary Gear Set with Helical Gears
- ODrive S1 FOC Controller with an Onboard Encoder
- 13x 3D Printed Parts
- Air Vents for Passive Cooling

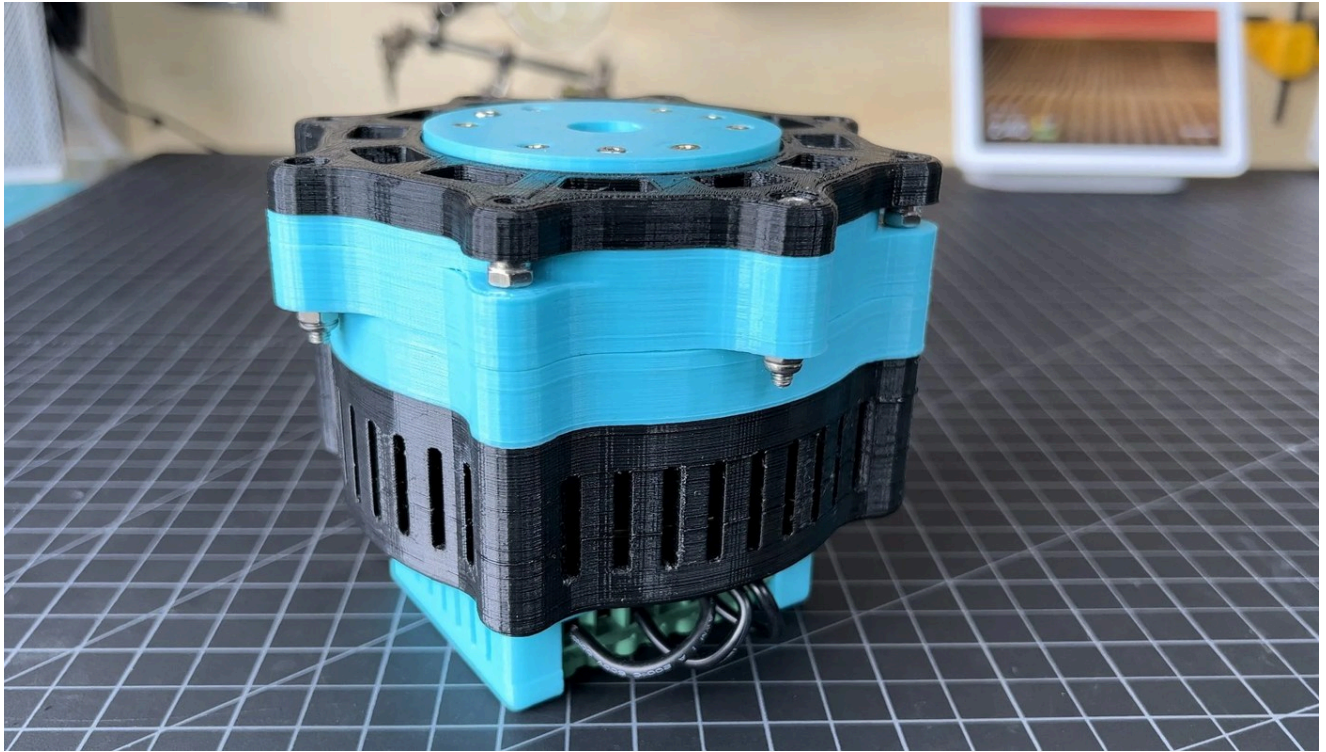
- Peak Holding Torque: 16.36 Nm
- Total Mass: 935g
- Total Cost: \$247



Quasi Direct Drive (QDD)

The QDD scheme was created by MIT and implemented in the Mini Cheetah. A QDD actuator is made with two key components.

- A brushless motor with a large gap radius to maximize torque density.
- A gear drive with a low gear reduction (generally between 3:1 and 10:1) for high torque output, high efficiency, and high backdrivability.



About the Design

The actuator uses a 90KV Eagle Power BLDC Motor as well as an ODrive S1 FOC controller. In closed-loop control mode, the S1 allows you to control the position, velocity, and torque of a brushless motor. With an onboard encoder, the S1 rests on the bottom of the actuator in order to measure the BLDC motor's position. The top of the actuator features a 9:1 planetary gear drive. I've designed helical gears for the gearbox since they are preferred for high-load applications and reduce noise. The actuator design also features the output bearing on the inside of the assembly meaning that OpenQDD doesn't need a gearbox cover to operate.

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Ball Balancer

Capstan Drive

출처: <https://www.aaedmusa.com/projects/capstandrive>



Capstan Drive (05/31/2024)

About

Gears are the common choice for a speed reducer in robotics, but I have a few points of contention. Gears have backlash which limits precision. Gearboxes can also be a bit loud especially if 3D printed. While it is possible to reduce backlash and noise with high-precision metal gears, purchasing such gears can be quite expensive. The solution to this is a rope-driven reducer called a Capstan Drive. I stumbled across this unique reducer when I discovered Stanley, a 12DOF quadrupedal robot by the company Ahead.IO. Considering that Capstan drives are not very popular, they quickly piqued my interest.

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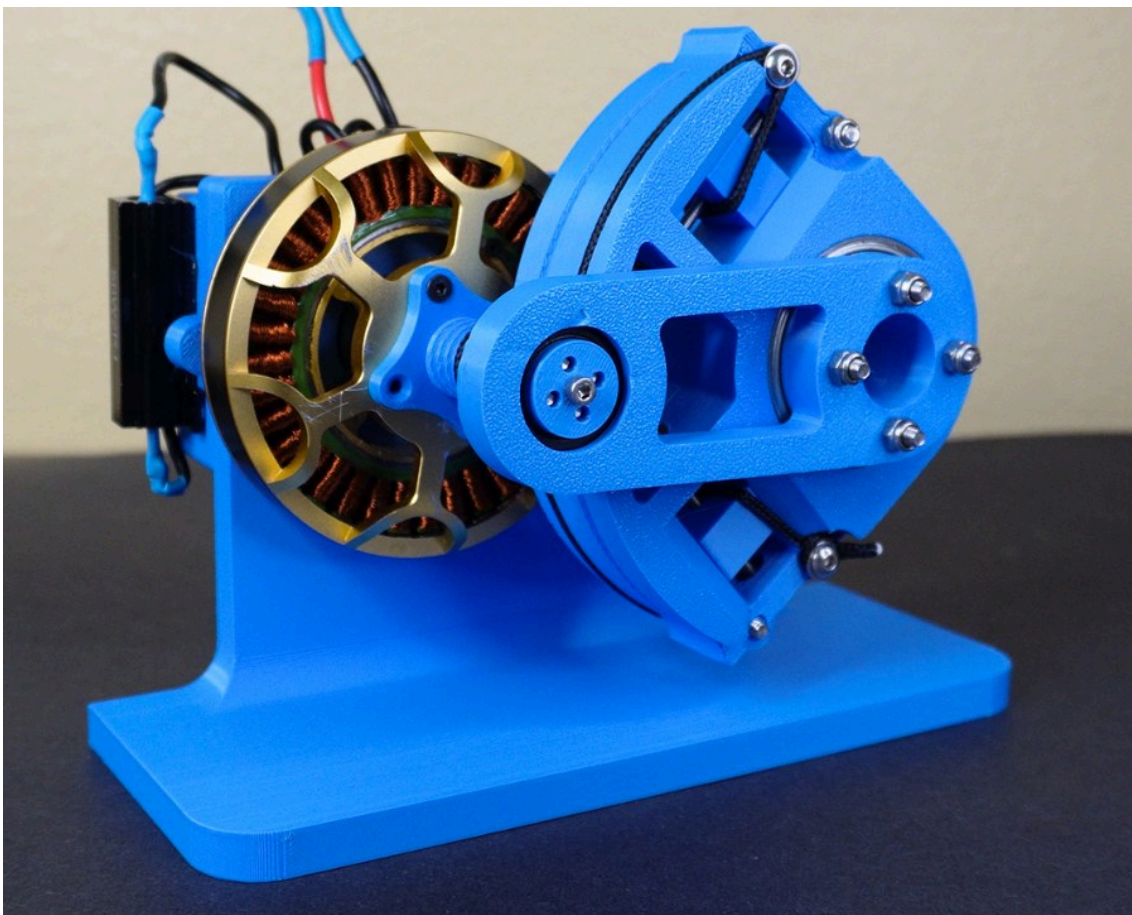
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Rope Selection

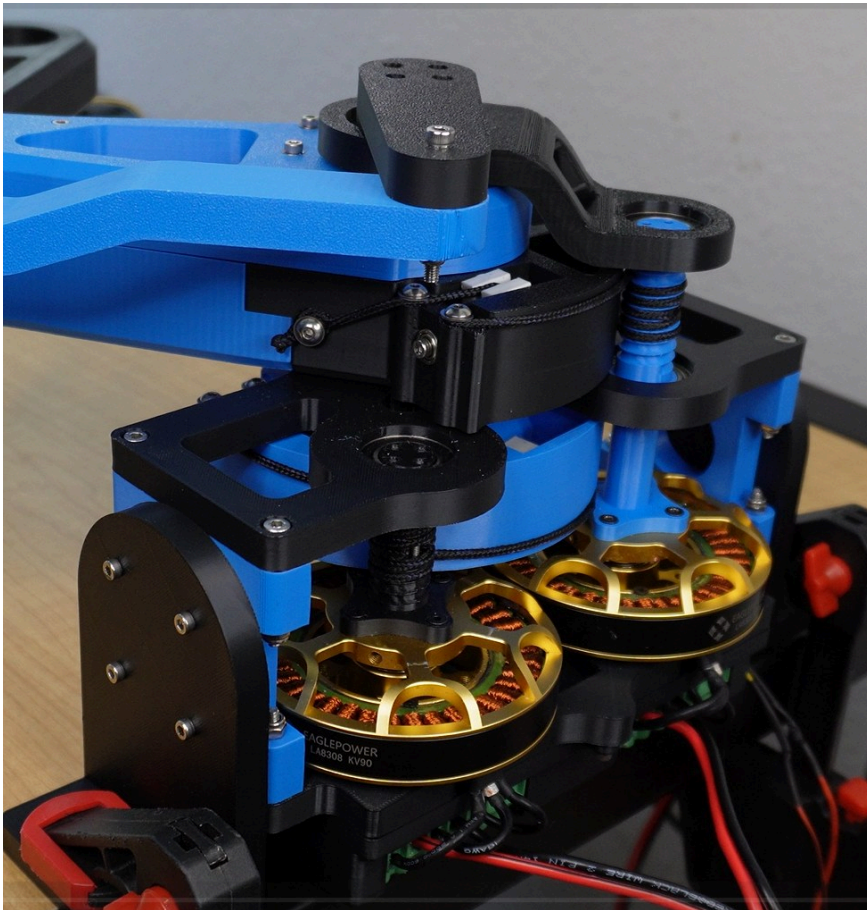
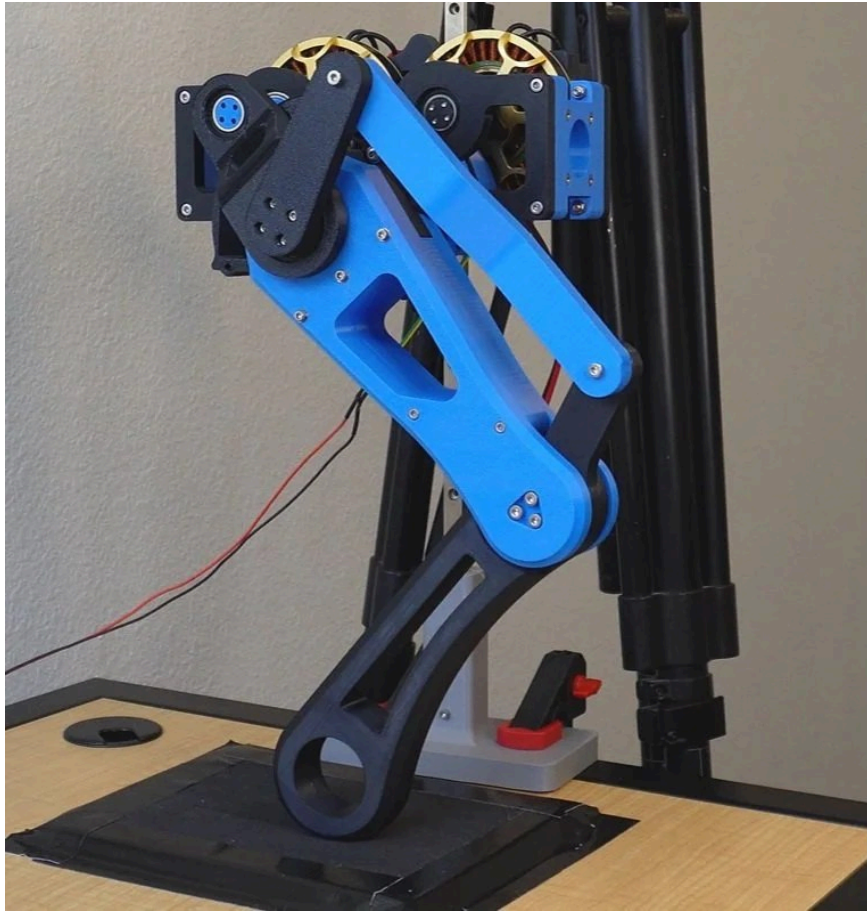
There is a lot of considerations that go into designing a capstan drive and the main one is rope selection. I started my rope selection journey with stainless steel wire which was a prospect short-lived. I had originally built a test actuator with the stainless steel wire and let it run continuously. After 4 short hours later, the wire broke. I ran two more trials and the wire continued to break each

time. None of the trials lasted more than 5 hours. This was quite surprising to me considering that the wire was extremely strong. After doing some digging I found that the cause of failure had nothing to do with strength and everything to do with a D/d ratio. In the case of a capstan drive, the D/d ratio is the ratio of the small drum diameter to the rope diameter. For metal wires, the minimum D/d ratio is quite high meaning that the small drum needs to be fairly large in diameter to prevent failure. I then moved my search to synthetic ropes which have much lower D/d ratios. As always in engineering, there was a tradeoff to this decision. With synthetic cable, I quickly discovered creep or the permanent stretch of rope. One important step in assembling a Capstan Drive is to ensure that the rope is fully tensioned to get rid of any slack. When put in tension, every rope will stretch. The only question is how long until it stretches significantly. When rope in a capstan drive undergoes creep, it goes out of tension and creates slack resulting in non-precise positioning. Unfortunately, most synthetic ropes have higher creep rates than what is needed for Capstan Drive. Since re-tensioning the rope every day seems quite inefficient, I conducted more research and discovered two ropes that claim to be inextensible: Vectran and Dyneema DM20. After performing a two-week endurance test I came to the conclusion that DM20 was the best rope for a Capstan Drive.



5R Parallel Robot

The second device that I built to test the capstan drive was a 5R parallel robot more commonly known as a 5-bar linkage. This robot has 2DOF and uses a coaxial design similar to the Stanford Doggo. The design of the robot uses the design of the test stand. I derived inverse kinematic equations for the robot and tested speed and precision by making it jump along a linear rail and draw basic shapes.



▶ 임베드: https://cdn.embedly.com/widgets/media.html?src=https%3A%2F%2Fwww.youtube.com%2Fembed%2FMwIBTbumd1Q%3Ffeature%3Doembed&display_name=YouTube&url=https%3A%2F%2Fwww.youtube.com%2Fw

How it Works

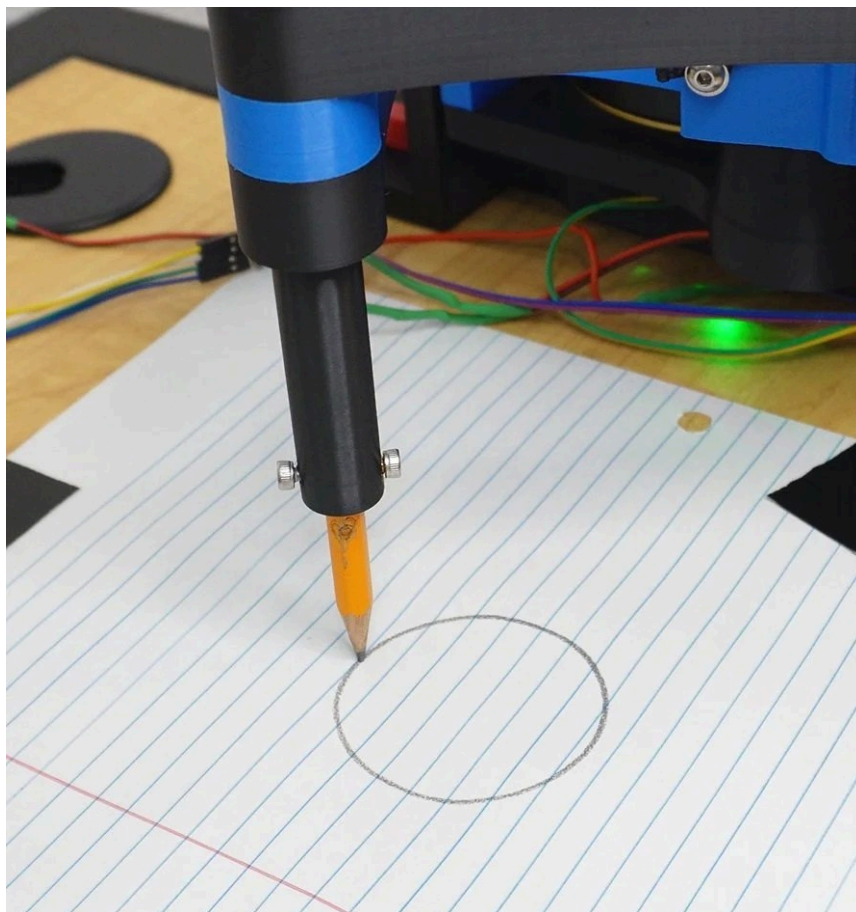
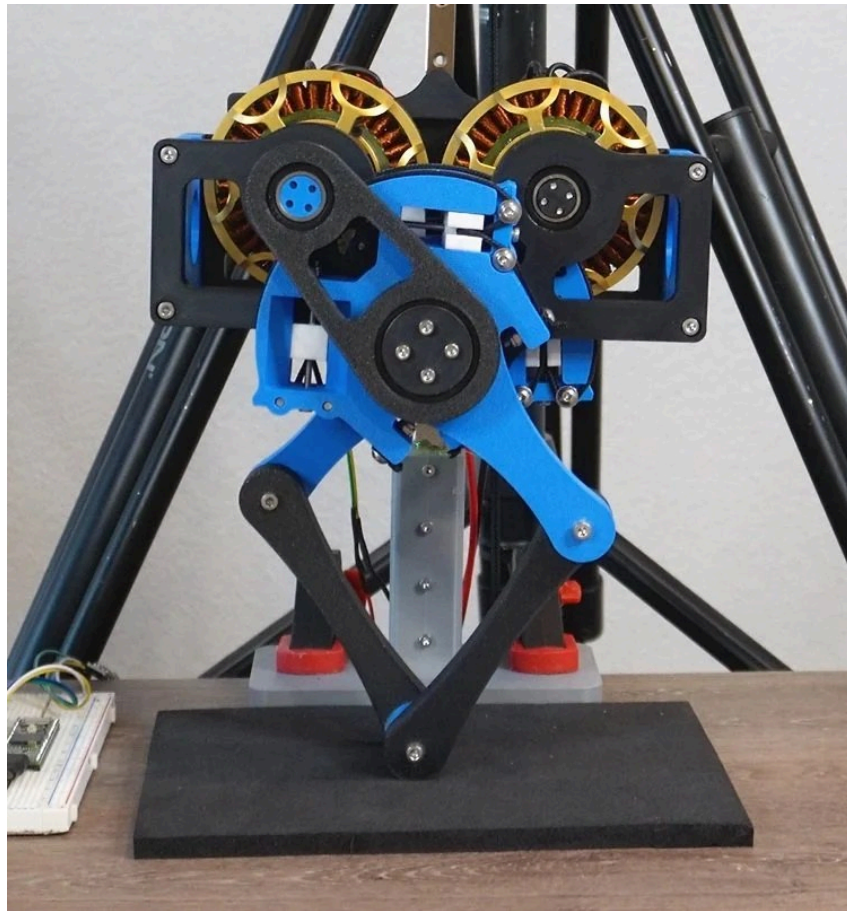
Imagine a small drum and a big drum where rope is wrapped around the smaller drum and fixed to the bigger drum on both sides. With this configuration, the drums become meshed together in the same way as a set of gears. Furthermore, the drums operate on the same principle of mechanical advantage that gears do. The gear reduction is equal to the ratio of the drum diameters. The benefits of a capstan drive include being relatively inexpensive, having zero backlash, low inertia, high torque transparency, and low noise. One question that I had when first exploring this reducer was “why doesn’t the rope slip if it’s just wrapped around the smaller drum?”. The answer to that question lies in the capstan equation. With each turn of rope on a drum, the amount of friction increases exponentially. With 3-5 turns of rope, there is enough friction for slipping to not be an issue. One of the obvious drawbacks of a Capstan Drive is limited rotation, but this isn’t an issue if your robot doesn’t need fully rotating joints i.e. quadrupeds and bipeds.

Actuator Test Stand

To test the effectiveness of a capstan drive I decided to build 3 devices. The first device is a robotic actuator. I went through lots of prototyping with the drum design, tensioning methods, and rope mounting options. I eventually settled on the design seen on the left which has the following features:

- 3D printed in PLA
- Weighs 852g
- 120° rotation
- 8.55:1 reduction (quasi-direct drive)
- Helical pattern on both drums for guided unraveling
- Lead screw to tension the rope
- ODrive S1 FOC Controller
- Eagle Power 90KV BLDC Motor

One of the issues that I encountered while designing the test stand was getting an exact gear reduction. I intended for the reduction to be 8:1, but for some reason making the big drum exactly 8 times the diameter of the smaller drum doesn’t do the trick. I believe this has to do with the helical pattern of the drums, but I am unsure of how to account for that in the gear reduction calculation. I ended up just leaving the reduction as 8.55:1 since its pretty close to the desired value. The test stand design is open source and can be downloaded on GitHub.



Quadrupedal Robot Leg

The third and final device that I built to test the Capstan Drive was a single 2DOF quadrupedal robot leg. I plan on building a robot dog with a capstan drive in the near future so I was curious to test the waters. The quad leg shares a very similar design to the 5R Parallel Robot since both use a coaxial shaft design. Just like with the 5-bar linkage, I derived inverse kinematic equations for the quad leg and made it jump along a linear rail. Overall the capstan drive is extremely smooth compared to a conventional gearbox. It's the exact reducer that I have been searching for to make affordable yet high-performing robots.

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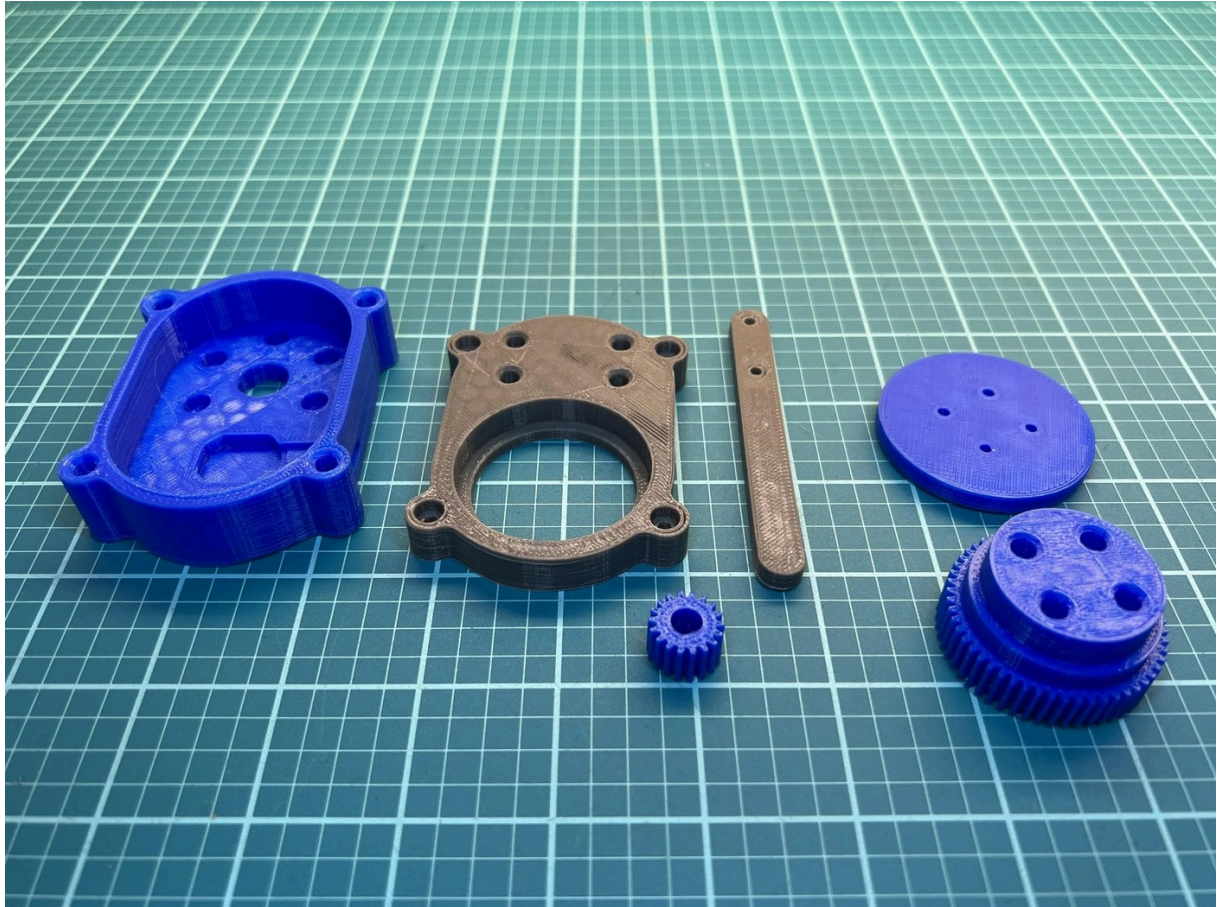
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3D Printed Servo

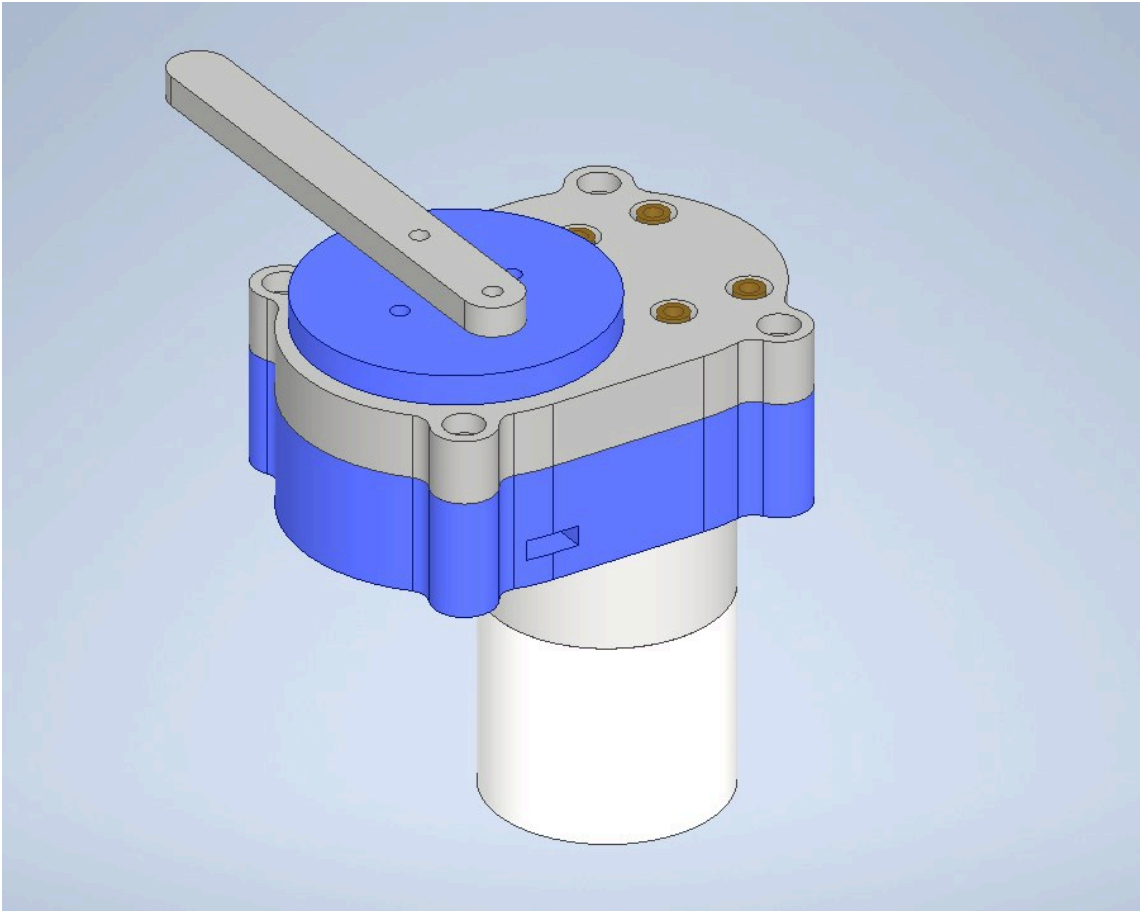
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3D Printed Servo (01/18/2023)



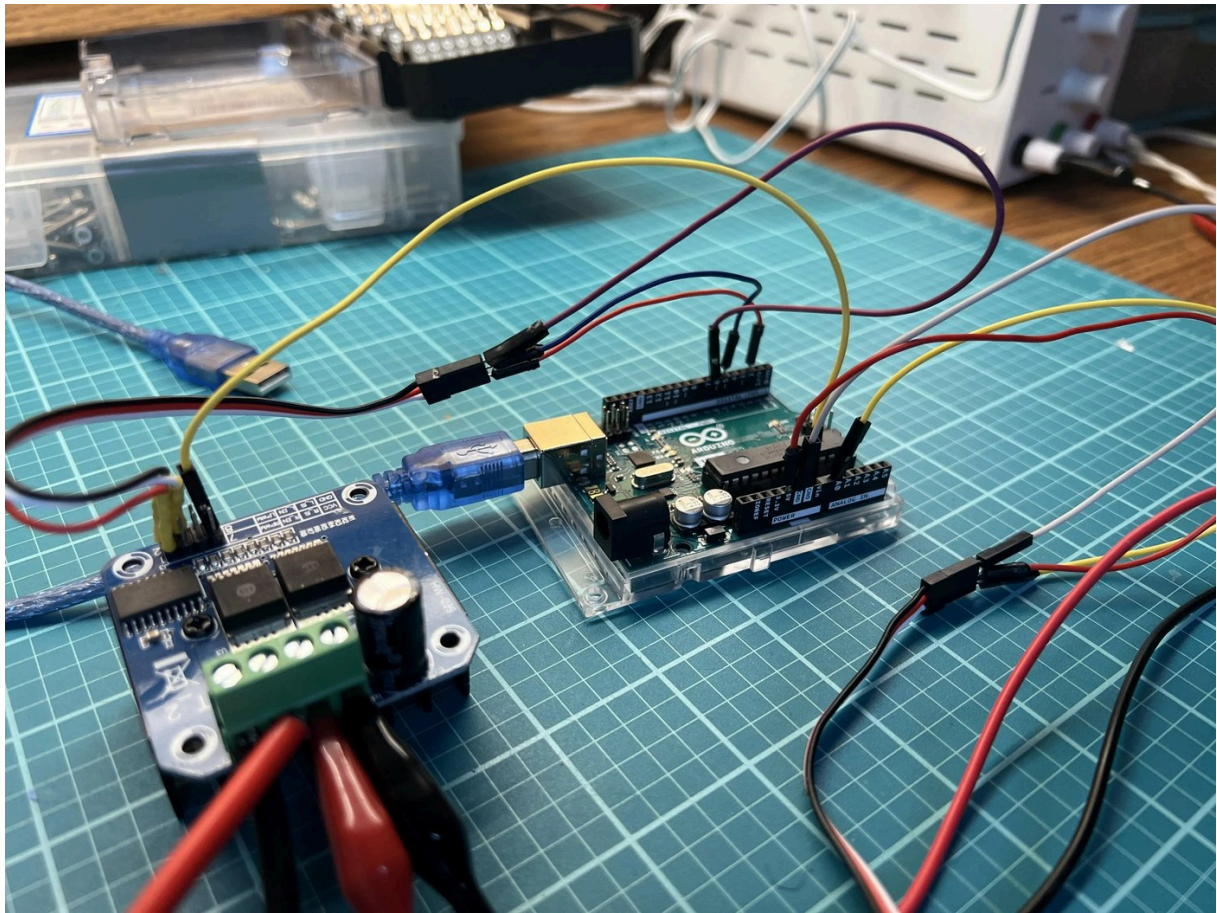
What?

- Created a 3D-printed servo motor with more applications than a hobby servo.
- Easy to produce via FDM printing.



How?

- Uses a cheap DC motor, an encoder, arduino uno, and motor driver.
- CAD in Autodesk Inventor



Results?

- 360 degrees of rotation
- Position feedback
- Easy to assembly
- Fully documented and open-sourced
- Could have more torque

▶ 임베드: <https://www.youtube.com/embed/BWSH7EnjaQE?feature=oembed>

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